Project Proposal – Team 03

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Project/Team Title: ZoneSort: Automated Parcel Sorting for HDB Estates

Introduction / Problem Statement:



Fig 1. Parcels sorted on a void-deck floor



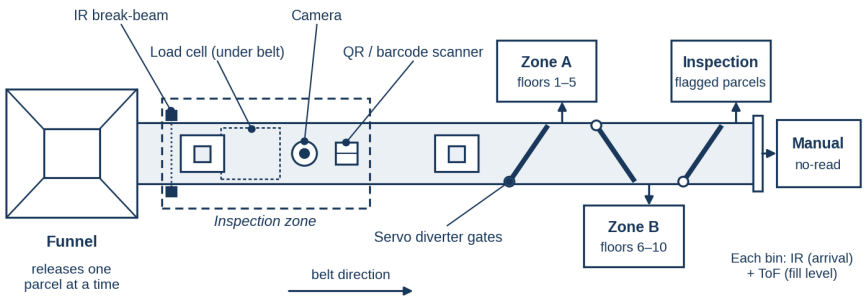
Fig. 2. SingPost POPStation parcel lockers

With the growing volume of e-commerce deliveries in Singapore, couriers often have to identify and sort large numbers of parcels by hand at residential estates. This sorting frequently happens on the floor of HDB void decks (Fig 1). Parcels are left exposed to the weather, recipients' personal details are visible, and packages are handled roughly. In one Boon Lay case, a courier sorted parcels at a void deck daily for about nine months, drawing repeated complaints from residents. Manual sorting takes time, and couriers may have little information about which blocks will require the most delivery capacity at different times.

Existing parcel-collection infrastructure, such as SingPost POPStation lockers (Fig 2) and residential smart lockers, supports parcel collection. Our proposal automates the sorting of incoming parcels by delivery zones, reducing the manual sorting couriers do before delivering parcels to individual units. It also uses the resulting traffic records to help supervisors plan courier allocation.

Proposed IoT Solution:

The proposed system is an IoT-enabled automated parcel sorting solution designed to reduce manual handling by delivery couriers. A courier loads parcels into a feeder magazine, and from there the system handles everything automatically. The system releases one parcel at a time, scans its code, obtains its zone assignment, and uses a conveyor and servo-controlled gates to route it. Unreadable or unknown labels go to a manual handling compartment. A local gateway maintains the zone mapping and communicates with the two ESP32 nodes; the cloud stores event records and forecasts which blocks are likely to receive higher parcel volumes to aid planning support for supervisors, who decide whether additional couriers are needed.



Each parcel goes through the following steps:

1. **Singulation.** Parcels are deposited and a servo-driven pusher ejects only the bottom parcel through a slot sized to approximately one parcel height. This ensures that parcels enter the conveyor one at a time.

   The parcel then enters an inspection zone where a camera positioned above the conveyor checks whether the shipping label is visible. If the label is detected, the parcel proceeds directly to the next stage. If no label is detected, a motorised flipping mechanism rotates the parcel and the camera checks it again.

2. **Identification.** An IR break-beam detects the parcel entering the inspection zone. A GM65 scanner reads its QR code or barcode to retrieve the corresponding blocks and floor information.
3. **Routing.** Prototype zones: A (floors 1-5), B (floors 6-10) and Manual (unreadable or unknown label). The decision is sent to the conveyor node over ESP-NOW. If the edge server does not reply in time, the parcel defaults to Manual.
4. **Sorting and confirmation.** Normal parcels are transported along the conveyor belt and directed to the appropriate zone compartment using servo-controlled sorting gates, while unreadable, unidentified parcels are diverted to the separate inspection area. The conveyor node tracks the parcel along the belt and fires the servo gate for its zone. An IR sensor at each bin confirms arrival.
5. **Logging.** Parcel information and system data, including block, floor and timestamps, are published over MQTT and stored in the cloud for monitoring, record-keeping, and long-term analysis.
6. **Forecasting and planning**. Data collected from multiple ZoneSort machines deployed across different residential blocks can be collated in the cloud. A forecasting model can then use this combined data to estimate parcel traffic for each block in the next planning period.

Flagged parcels refers to parcels that have damaged/unreadable labels

Sensors /Actuators/Hardware Used:

IoT Device 1	Component	Function
ESP32 1 (Parcel Inspection Node)	1. IR / Proximity Sensor 2. Barcode / QR Scanner 3. Load Cell + HX711 4. Camera Sensor 5. Time of Flight Sensor	1. Detects the arrival of a parcel 2. Retrieves parcel's block and floor infomation 3. Measures parcel weight/ detect whether an unusually heavy parcel has entered the system. 4. Detect/sense the label's orientation 5. Estimate dimensions of parcel

IoT Device 2	Component	Function
ESP32 2 (Conveyor & Sorting Node)	1. DC Motor 2. Motor Driver 3. Servo Motors 4. IR Sensors 5. Buzzer / LEDs	1. Drives the conveyor belt 2. Controls the conveyor motor 3. Controls the sorting gates to direct parcels 4. Detects parcel position along the conveyor 5. Provides system status and warning indications

A 3D-printed conveyor belt will be designed and fabricated to transport parcels through the singulation, inspection, and sorting stages of the system. The conveyor structure, rollers, and mounting components will be custom-designed to suit the prototype dimensions and parcel sizes.

Machine Learning Models:

Each successfully scanned parcel generates a timestamped record containing its destination block, floor range, scan outcome, and confirmed sorting outcome. The cloud aggregates these records into a daily count for each block, including days with zero parcels. The prediction target is the number of parcels expected for each block on the next delivery day. Inputs include block ID, day of week, the previous day's count, the count on the same weekday in the previous week, and rolling averages over the preceding 7 and 28 days. Only data available before the forecast is issued are used as inputs.

We will compare two ML models against a simple baseline that predicts the average volume for the same weekday: Poisson regression, which models non-negative parcel counts, and XGBoost regression, which can learn nonlinear

relationships between recent traffic, block, and day of the week. The models will be trained on earlier days and tested on later days. We will measure mean absolute error in parcels per block per day and whether each model correctly identifies the busiest blocks. The cloud dashboard displays these forecasts to help supervisors plan courier allocation; sorting and gate control continue locally without waiting for the forecast. Our project applies the forecasting task to the block-level records produced by an IoT sorting prototype.

Other Cool Features:

1. Dashboard: Current parcel counts and sorting outcomes by block, unreadable-label count, conveyor faults, historical traffic, and forecast counts for the next planning period.
2. Alerts: Notifications for jams or repeated scan failures
3. Fail-safe routing: Unknown labels and timed-out mapping requests go to Manual; an uncertain tracking state stops the belt for operator review.
4. Custom 3D-printed build: Feeder guides, conveyor mounts, gates, and collection compartments sized to the prototype parcels.

Progress so far:

The system concept, zone-level sorting workflow, two ESP32 roles, gateway responsibilities, and candidate hardware have been identified. The next steps are to finalise the feeder and conveyor design, define the block label format and local mapping, and design a realistic traffic dataset and evaluation plan.

Possible Limitations or Challenges:

Challenge/ Limitation	Mitigation
Unreadable parcel labels. Barcode or address may be blocked, damaged or poorly scanned.	Use standardised labels and controlled lighting; unreadable parcels are sent to manual handling.
Mechanical jams or sorting errors. Parcels of different sizes and weights may jam the conveyor or miss the correct gate.	Limit parcel size and weight for the prototype, and use a simple guide/funnel or controlled parcel spacing to prevent jamming.
Parcel tracking mismatch. Multiple parcels close together may cause destination data to be matched to the wrong parcel.	Assign each parcel a temporary ID (in the form of a barcode/image) and maintain sufficient spacing between parcels.
Limited data for traffic prediction. The prototype may not collect enough historical data to accurately predict future parcel demand.	Records from the physical prototype form the real dataset; we augment them with simulated traffic in the same format and evaluate the models on real prototype runs.

Timeline:

Week	Milestone
1	Finalise system design; test ESP32 boards, barcode scanning, motor and servo components
2	Build the conveyor and simple parcel guide/funnel; test DC motor and servo sorting gates
3	Implement barcode identification and map parcel destinations to Zones A/B/Manual
4	Integrate parcel scanning with conveyor sorting; test reliable parcel movement and correct gate activation
5	Set up MQTT/HTTP communication, cloud database and basic dashboard for parcel counts by zone/time
6	Implement simple parcel-traffic analytics and forecasting using collected or simulated historical data
7	Full system integration and testing; measure sorting accuracy, scan success rate and dashboard/forecast output; prepare demo